Adrenal Hemorrhage During Anticoagulant Therapy

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ADRENAL HEMORRHAGE is a rare complication of anticoagulant therapy which usually brings the patient to death before the diagnosis is made and treatment established. Review of the literature reveals that in the great majority of the reported cases, the diagnosis was an unsuspected finding at postmortem examination;[1–4,6,9,11–13,15] we have been able to find in the world literature only 3 cases in which the diagnosis was established *in vivo*. It is interesting that all 3 patients survived with appropriate treatment.[5,7,16]

To increase the clinical awareness in this catastrophic complication, we are reporting the case history of a patient in whom the diagnosis of acute adrenal insufficiency due to hemorrhage during anticoagulant therapy, was established by clinical and laboratory findings, and who survived with medical treatment.

Case Report

A 79–year–old Sicilian female had a right radical mastectomy for carcinoma of the breast. Pathological examination of the specimen revealed no residual tumor at the line of the excisional biopsy; all the lymph nodes were also clear. She did well until the fourth postoperative day when she developed a sudden episode of left chest pain; at physical examination the only positive finding was a slight tenderness in the right calf. The chest X-ray revealed exaggeration of bronchovascular markings of the left base. The following day, the tenderness in the right calf had increased and anticoagulation was instituted with heparin, 75 mg. q.6 hrs. subcutaneously. On the fourth day of anticoagulation, a bleeding time of 38 minutes was obtained and one dose of heparin was withheld. The following morning moderate bloody drainage was observed on the dressing of the mastectomy; the patient complained of back pain; anticoagulation was continued. The following night, now the sixth day of anticoagulation, the patient developed an acute and severe exacerbation of the back pain, mainly at the level of D11 and D12; she was pale, dyspneic and nauseated; her temperature which had been normal had risen to 101°, the heart rate was 110/m. and the blood pressure fell from 160/70 to 116/50mm. Hg. The abdomen was distended and there was some tenderness in the epigastrium, rectal examination revealed normal colored stools. Hemoglobin was 5.5 gm. and hematocrit was 19. The clinical picture closely resembled the one seen in a ruptured aortic aneurysm, but no pulsating masses had been felt in the abdomen on admission and none could be felt now; a radiogram of the abdomen revealed a calcified aorta of normal configuration. No clear diagnosis could be established; because of an old history of dyspepsia a gastrointestinal bleeding was suspected: anticoagulation was suspended and whole blood was transfused. The following morning the hemoglobin had returned to above 10 gm., the abdomen remained soft but distended and again normal colored stools were obtained at rectal examination. The blood pressure, however, remained unstable, the temperature had risen to 103° and the patient was very drowsy and in a few hours became lethargic. With this clinical picture, a massive adrenal hemorrhage was strongly suspected and after drawing blood for laboratory confirmation, treatment was instituted with intravenous normal saline and hydrocortisone, further blood transfusions and antibiotics. The laboratory, during the next 24 hours, submitted the following results: Blood sugar—58 mg.; BUN—36mg; CO_2—22; sodium—126; chloride—88; potassium—3.2 meq; eosinophil count—250/mm^3 (normal up to 50).

After 24 hours of treatment the condition of the patient dramatically improved. In the following days, the cortisone dose was reduced from the initial 400 mg. per/day I.V. and then substituted with Prednisone p.o.; fluorocortisone was also added. The

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patient progressed well and was discharged three weeks after the acute episode, on replacement therapy. Before discharge, 24-hour urinary 170H were 1.4mg. and 17K were 1 mg.; the skin was noted to have taken on a definite dark discoloration.

Twice in the months following discharge, the patient was readmitted for acute adrenal insufficiency. On both occasions the patient had failed to take the Prednisone tablet, once because she was nauseated and once when she went away on a weekend trip and forgot to take along her medication. Both times she presented with high temperature, drowsiness and low blood pressure; both times she was prompty revived with normal saline and cortisone intravenously. Last year the patient developed an acute myocardial infarction from which she recovered. At present she is 3 years after the radical mastectomy without evidence of recurrence of the breast carcinoma and on continous Prednisone medication.

Discussion

The possibility of an adrenal hemorrhage in this patient was first considered 24 hours after the onset of the severe acute episode of back pain and shock. This suspicion was based on 3 clinical findings: the consideration that blood pressure remained unstable despite adequate blood replacement; the rising temperature in the absence of infection, and the observation that the patient in a short time became drowsy and then lethargic.

In retrospect, the severe back pain which ushered the catastrophe, should have alerted us to this possibility once the diagnosis of ruptured aortic aneurysm had been ruled out; however, this important diagnostic element was unfortunately underestimated and caused a 24-hour delay in considering the proper diagnosis.

Amador[1] in 1965, published a comprehensive review of all the recorded cases of adrenal hemorrhage occuring during anticoagulant therapy. He collected 20 cases from the world literature and added 10 cases from his own institution. In three of the collected cases, the diagnosis was made *in vivo*.[5,7,16] We have been able to collect 19 more cases: in all these patients the diagnosis was made at autopsy.[2–4,6,9,11–13,15] Another case with survival was published by Planques[14] in 1959. In reading this case history one wonders if it could not have been the same case published by Decourt[5] in 1960. In a case reported by Klassen[8] adrenal insufficency of gradual onset occurred after an episode of retroperitoneal hemorrhage associated with anticoagulant therapy.

This complication has been observed both with heparin and coumadin treatment; in many of the published cases, anticoagulation was well beyond theapeutic range before the catastrophic event. Almost invariably, the presenting sign is severe back or flank pain, acute in onset, associated with anorexia, nausea, vomiting and frequently with temperature elevations. Hypotension and impairment of the sensorium follows, the patient may become lethargic, and if corrective treatment is not instituted, almost invariably dies in a short period of time. The laboratory findings reveal a low glucose, low sodium and low chloride and elevated potassium and BUN values. The eosinophil count in those few cases where it was measured was elevated above 50/mm.[3]

The pathogenesis of the hemorrhage in the adrenal parenchyma is not clear, probably it is mainly the consequence of the abnormal coagulation state. The prevalent old age of the published patients, with its attendant increased vascular fragility, and the stress condition arising from the medical or surgical condition for which anticoagulation was required might also play a role since it is known that adrenal hemorrhage may occur spontaneously under stress.[10] Experimentally, it has been shown that ACTH when added to anticoagulants in rabbits increases ten-fold the incidence of hemorrhagic complications.[17]

Judging from the results obtained in the few cases where diagnosis was made, including the present one, the treatment should consist in discontinuation of anticoagulation and generous replacement with steroid, sodium chloride and blood. This medical replacement treatment has been effective in the few cases where it was initiated before coma occurred; however, since usually there is little room for delay, it should be instituted on a clinical suspicion without waiting for laboratory confirmation.

Summary

A patient after radical mastectomy developed deep thrombophlebitis of the leg and was treated with heparin. On the sixth day of anticoagulation she developed a sudden episode of shock, hyperthermia and lethargy. An adrenal hemorrhage was suspected and the patient was treated accordingly with prompt recovery. The laboratory findings confirmed the clinical diagnosis.

References

1. Amador, E.: Adrenal Hemorrhage During Anticoagulant Therapy. Ann. Intern. Med., **63**:559, 1965.
2. Bianchi, C., Camerini, F., Crepaldi, L., *et al.*: Pathologia da anticogulanti. Minerva Cardioangiol., **18**:316, 1970.
3. Castleman, B. and Kibbee, B. U.: Case Records of the Massachusetts General Hospital. New Eng. J. Med., **267**:1140, 1962.
4. Coury, C. H., Thiboult, P. H. and Bourdarias, J. P.: Hematome bilateraux des capsules surrenales Côeur. Med. Int., **4**:15, 1965.
5. Decourt, J. and Michard, J. P.: Insuffisance surrénale aigüe, puis chronique installée brutalement au cours d'un traitement par l'héparine. Bull Soc. Med. Paris., **76**:422, 1970.

6. Grivâux, M., Diallo, A., Barbier, J., *et al.*: La nécrose hémoragique des surrenales. Sem. Hôp. Paris., **46**:2507, 1970.

7. Harper, J. R., Ginn, W. M. and Taylor, W. J.: Bilateral Adrenal Hemorrhage–Complication of Anticoagulant Therapy, Case Report and Review of the Literature. Amer. J. Med., **32**:984, 1962.

8. Klassen, J. and Murphy, B. E.: Survival After Bilateral Adrenal Hemorrhage During Heparin Therapy. Can. Med. Assoc. J. **97**:1162, 1967.

9. Lansing, P. F.: Adrenal Hemorrhage Associated with Dicumarol Anticoagulation. J. Maine Med. Assoc., **52**:207, 1961.

10. Lawson, D. W., Corry, R. J., Patton, A. S., *et al.*:Massive Retroperitoneal Adrenal Hemorrhage, Surg. Gynecol. Obstet., **129**:989, 1969.

11. Leblanc, M., Michiels, R., Beaufils, J. P., *et al.*: Les Hématomes bilaterâux des glandes surrénales, Complication du traitment anticoagulant. Sem Hôp Paris, **44**:302, 1968.

12. Markle, C. D. and Evans, E. C.: Adrenal Hemorrhage Associated with Anticoagulant Therapy. South. Med. J., **63**: 526, 1970.

13. McDonald, F. D., Myers, A. R. and Pardo, R.: Adrenal Hemorrhage During Anticoagulant Therapy. JAMA, **198**:1052, 1966.

14. Planques, J., Santmarc, R. and Chisin, E.: Insuffisance surrénale apres' héparinotherapie. Annal Méd Légale, **39**: 266, 1959.

15. Reece, R. L.: Bilateral Adrenal Hemorrhage and Calcification of Acute Myocardial Infarction, Report of a Case on Anticoagulant and Steroid Therapy, Hartford Hosp. Bull., **19**: 24, 1964.

16. Thorn, G. W., Goldfien, A. and Nelson, D. H.: The Treatment of Adrenal Dysfunction. Med. Clin. N. Amer., **40**:1261, 1956.

17. Van Couwenberge, G. H. and Jaques, L. B.: Hemorrhagic Effect of ACTH with Anticoagulants. Can. Med. Ass. J., **79**:536, 1958.